

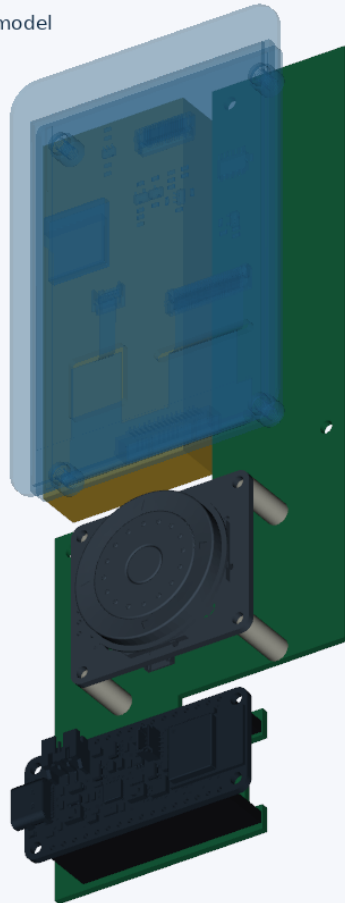
From size dummy to routed board

Outside size accepted: **74 x 178 x 28 mm**. USB first; battery space reserved.

HARMONY / MARK 6

Component stack review

74 x 178 x 28 mm Waveshare 29318 vendor model



Actual vendor Feather, display and ANO models. Cable paths and final housing need physical fit checks.

CAD checks passed. Zero schematic ERC violations, zero PCB DRC violations and zero open connections. All 122 electrical PCB pads match the schematic; all 28 Feather mating positions match vendor CAD.

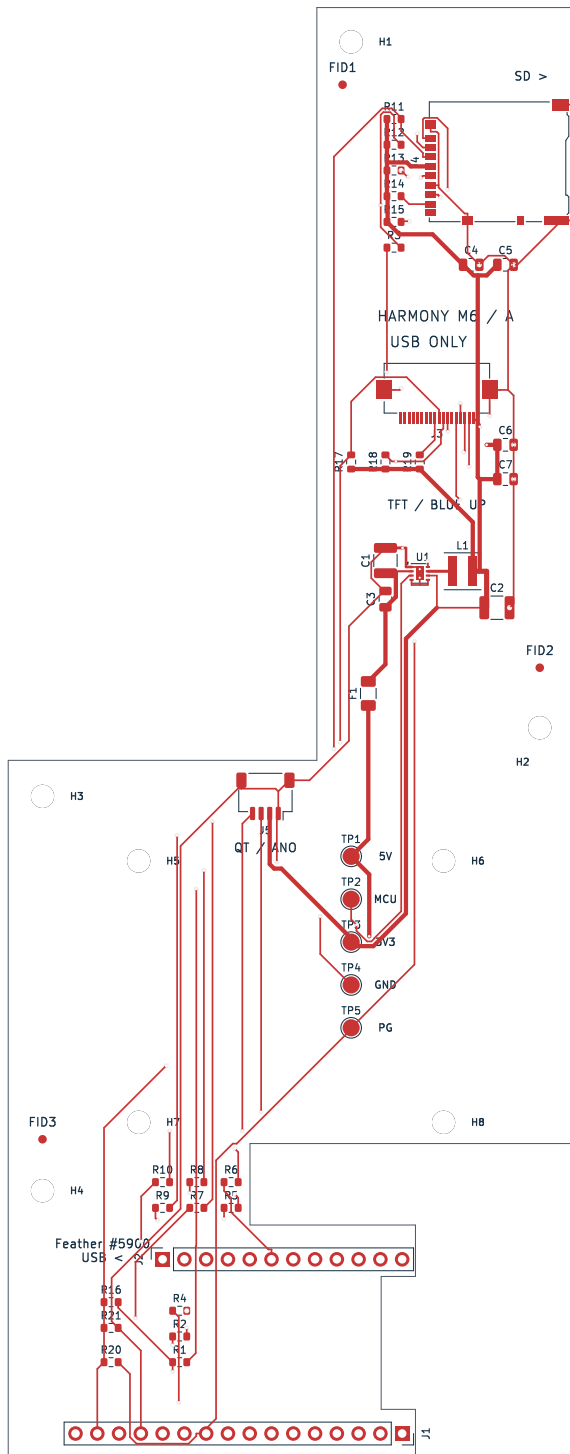
No hand soldering. The supplier fits 36 carrier parts, including the sockets. The factory-headered Feather, EYESPI display and ANO controls plug in.

Remaining physical checks. Exact display thickness, factory header insertion and cable/port access. Waveshare 29318 replaces #5846; US stock is listed. Physical module and cable fit remain unverified.

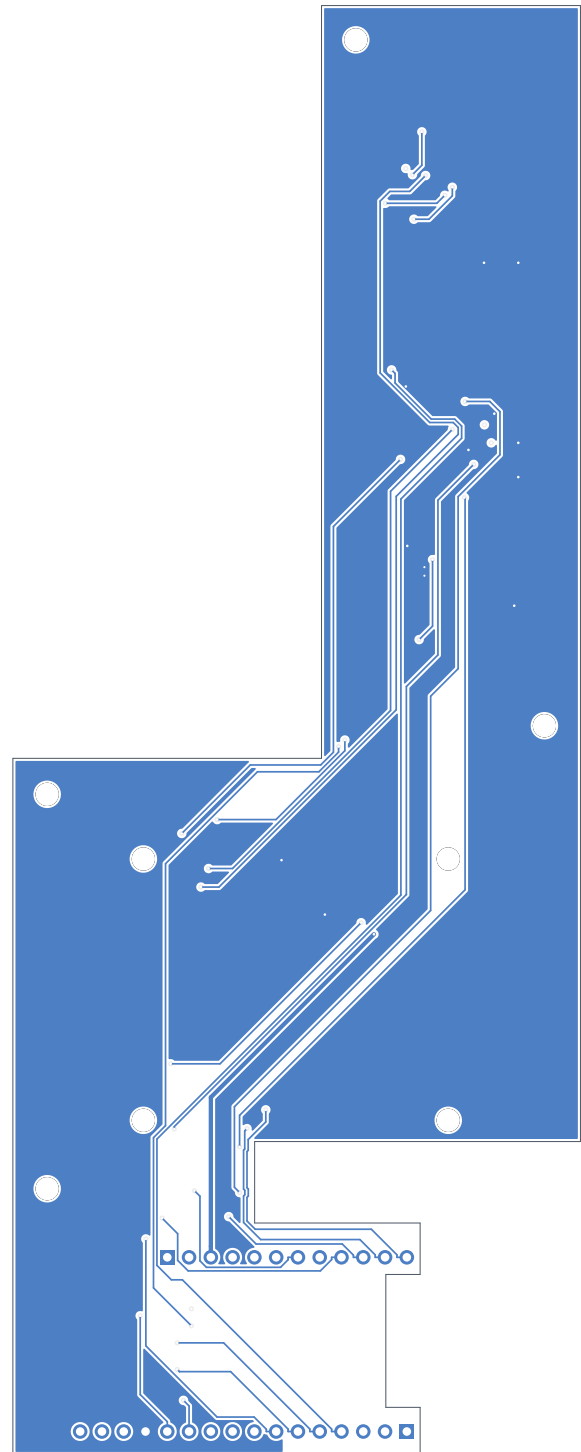
Routed carrier and antenna cutout

Four layers, 1.6 mm board. Top and back shown from the same top-view coordinates.

TOP / COMPONENT SIDE



BACK / THROUGH VIEW



The antenna overhangs the carrier cutout. Ground pours stop outside its reserved halo; necessary header pads and traces remain nearby. Final in-case Bluetooth range still needs testing.

Assembly stack and release checks

All heights below are measured from the rear outside surface toward the front.

Item	Height / Z range	Basis
Carrier PCB	6.5-8.1 mm	1.6 mm board
Socket body top	16.61 mm	Samtec SSW: 8.51 mm
Feather highest component	25.52 mm	Vendor STEP; nominal headers
ANO control front	27.47 mm	Vendor STEP; 12 mm spacers
Waveshare display	16.75-27.3 mm	Vendor STEP; 10.55 mm total
Future battery reservation	1.6-12.1 mm	Unconnected; no charger

Power

TPS62162 with 2.2 uH and 22 uF / 16 V bulk capacitors. A 450 mA peripheral allowance plus the Feather gives about 618 mA from a 4.75 V source at assumed 85% converter efficiency. Use a known 5 V source with at least 1 A available for qualification. These are budgets, not measurements.

Cables and case ports

100 mm EYESPI cable (#5239) and 50 mm QT-to-QT cable (#4399). The display FFC connector sits near the top, at X37/Y20.3. Reserve a loose cable loop above the future battery and use the upper ANO connector. Provide recessed USB plug access and a finger pocket for SD ejection.

Before an order

Supplier DFM, exact part availability, header/socket fit and the display measurement remain. Vias inside SMT pads require resin filling and copper capping; the assembler must review U1 stencil coverage. Manufacturing files are supplied for review, with no order authorization.

After assembly

Stage first power with current limiting and rail checks. Then test display refresh, SD and controls during A2DP playback, reconnect behavior, temperature and RF range. Complete the deferred full-album run before calling Mark 6 qualified.

Detailed sources, exact BOM, tolerances and procedures are in the accompanying power-review.md and mechanical-review.md. The STEP assembly is a review model, not a finished hollow case.